

Important Questions

Applied Physics for Computer Science

Stars indicate how often each question appeared in exams:

★★★★★ = 5 times ★★★★☆ = 4 times ★★★☆☆ = 3 times

Module 1: Lasers & Fiber Optics

1. Define acceptance angle and numerical aperture, and derive an expression for numerical aperture in terms of the refractive indices of core and cladding.

(Appeared in VTU exams: Jan 2023, June 2023, June 2024, Jan 2025)

See Solved Model Question Paper 01, Question 1(b), Page 3

★★★★☆

2. Discuss the types of optical fibers based on modes of propagation and refractive index profile.

(Appeared in VTU exams: Jan 2023, June 2023, Jan 2024, June 2024, Jan 2025)

See Solved Model Question Paper 01, Question 2(b), Page 6

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3. Illustrate the construction and working of a semiconductor LASER with a neat sketch and energy level diagram, and mention its applications.

(Appeared in VTU exams: Jan 2023, June 2023, June 2024, Jan 2025)

See Solved Model Question Paper 01, Question 2(a), Page 5

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4. Obtain the expression for energy density using Einstein's A and B coefficients and thus conclude that $B_{12} = B_{21}$.

(Appeared in VTU exams: Jan 2023, June 2023, Jan 2024, June 2024, Jan 2025)

See Solved Model Question Paper 02, Question 1(a), Page 1

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Module 2: Quantum Mechanics

1. State and explain Heisenberg's uncertainty principle. Show that an electron cannot exist inside the nucleus using Heisenberg's uncertainty principle.
(Appeared in VTU exams: Jan 2023, June 2023, Jan 2024, June 2024, Jan 2025)
See Solved Model Question Paper 02, Q4(b), Page 13 & Model QP 01, Q3(b), Page 10

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2. Set up the Schrödinger time-independent wave equation in one dimension.
(Appeared in VTU exams: Jan 2023, June 2023, Jan 2024, June 2024, Jan 2025)
See Solved Model Question Paper 01, Question 3(a), Page 9

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3. Discuss the motion of a quantum particle in a one-dimensional infinite potential well of width 'a'. Also, obtain the eigenfunctions and energy eigenstates for a particle in the box (infinite potential well).
(Appeared in VTU exams: Jan 2023, Jan 2024, June 2024, Jan 2025)
See Solved Model Question Paper 01, Question 4(a), Page 11

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Module 3: Quantum Computing

1. Elucidate the differences between classical and quantum computing.
(Appeared in VTU exams: Jan 2023, June 2023, Jan 2024, June 2024)
See Solved Model Question Paper 01, Question 6(b), Page 18

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2. State the Pauli matrices and apply them to the states $|0\rangle$ and $|1\rangle$.
(Appeared in VTU exams: Jan 2023, Jan 2024, June 2024, Jan 2025)
See Solved Model Question Paper 01, Question 6(a), Page 17

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3. Discuss the CNOT gate and its operation on four different input states.
(Appeared in VTU exams: Jan 2023, June 2023, Jan 2024, June 2024, Jan 2025)
See Solved Model Question Paper 01, Question 5(b), Page 16

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4. Explain the representation of a qubit using the Bloch sphere.
(Appeared in VTU exams: Jan 2023, Jan 2024, Jan 2025)
See Solved Model Question Paper 02, Question 6(a), Page 16

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Module 4: Electronic Properties of Materials

1. Enumerate the failures of classical free electron theory and the assumptions of quantum free electron theory.

(Appeared in VTU exams: Jan 2023, Jan 2024, June 2024, Jan 2025)

See Solved Model Question Paper 02, Question 7(a), Page 19

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2. Describe the Meissner effect and classify superconductors into Type I (soft) and Type II (hard) using M-H graphs.

(Appeared in VTU exams: Jan 2024, June 2024, Jan 2025)

See Solved Model Question Paper 01, Question 8(a), Page 23

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3. Define the Fermi factor and discuss its variation with temperature and energy.

(Appeared in VTU exams: Jan 2023, June 2023, Jan 2024, June 2024, Jan 2025)

See Solved Model Question Paper 01, Question 7(a), Page 19

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4. Explain superconductivity and discuss BCS theory for negligible resistance near absolute zero.

(Appeared in VTU exams: June 2023, Jan 2024, Jan 2025)

See Solved Model Question Paper 02, Question 8(a), Page 22

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Module 5: Animation & Statistics

1. Describe jumping and its parts.

(Appeared in VTU exams: Jan 2023, Jan 2024, June 2024, Jan 2025)

See Solved Model Question Paper 01, Question 10(a), Page 27

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- 2.** Discuss timing in Linear motion, Uniform motion, slow in and slow out.

(Appeared in VTU exams: Jan 2023, June 2023, Jan 2024)

See Solved Model Question Paper 01, Question 9(a), Page 25



- 3.** Discuss modeling proton decay probability using Poisson distribution.

(Appeared in VTU exams: Jan 2023, June 2023, Jan 2025)

See Solved Model Question Paper 02, Question 10(b), Page 28



- 4.** Discuss features of normal distribution using bell curves.

(Appeared in VTU exams: June 2023, Jan 2024, Jan 2025)

See Solved Model Question Paper 01, Question 10(b), Page 28

